The Unreasonable Effectiveness of Mathematics in Physics

HiSEP Seminar 2026

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In 1960, Eugene Wigner marveled that mathematics is unreasonably effective

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The Unreasonable Effectiveness of Mathematics in the Natural Sciences

Richard Courant Lecture in Mathematical Sciences delivered at New York University, May 11, 1959

EUGENE P. WIGNER
Princeton University

"and it is probable that there is some secret here which remains to be discovered." (*C. S. Peirce*)

Let me end on a more cheerful note. The miracle of the appropriateness of the language of mathematics for the formulation of the laws of physics is a wonderful gift which we neither understand nor deserve. We should be grateful for it and hope that it will remain valid in future research and that it will extend, for better or for worse, to our pleasure even though perhaps also to our bafflement, to wide branches of learning.

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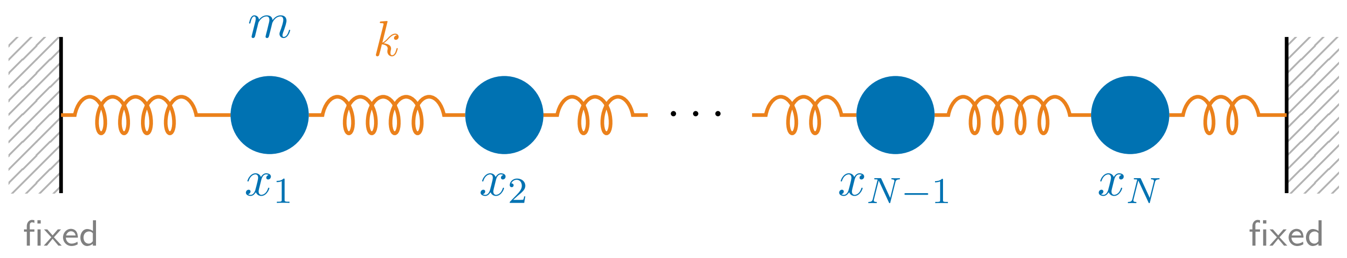
SECTION 1

Classical Mechanics

古典力学

The system: N masses coupled by springs



Newton's law（ニュートンの運動方程式） for the displacement（変位） $x_j(t)$ of mass j (walls: $x_0 = x_{N+1} = 0$):

$$m\,\ddot{x}_j \;=\; k\,(x_{j+1} - x_j) - k\,(x_j - x_{j-1}) \;=\; -\,k\left(2x_j - x_{j+1} - x_{j-1}\right)$$

Each mass feels only its *two neighbours* — the pattern $(-1,\,2,\,-1)$. Collecting the displacements into $\boldsymbol{x} = (x_1, \ldots, x_N)^{\mathsf{T}}$, that pattern becomes the rows of one matrix:

$$\ddot{\boldsymbol{x}} \;=\; -\frac{k}{m}\,A\,\boldsymbol{x}, \qquad A \;=\; \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & \ddots & \\ & & \ddots & \ddots & -1 \\ & & & -1 & 2 \end{pmatrix}$$

- The problem reduces to finding *eigenvectors*（固有ベクトル） $\boldsymbol{v}$ and *eigenvalues*（固有値） λ:

$$A\boldsymbol{v} = \lambda \boldsymbol{v}$$

- **Assumption:** a sampled sine wave, $v_j = \sin(j\varphi)$.

- Using $\sin\big((j{+}1)\varphi\big) + \sin\big((j{-}1)\varphi\big) = 2\cos\varphi\,\sin(j\varphi)$:

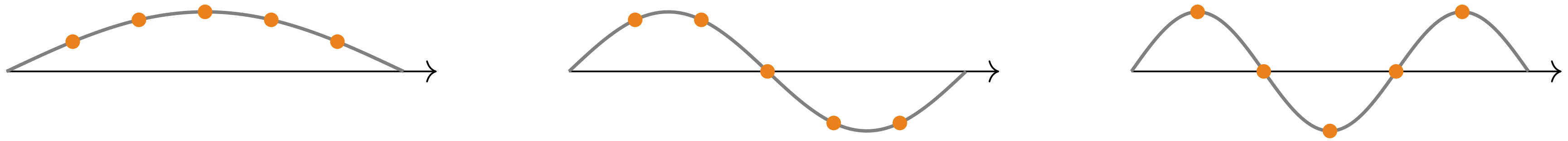
$$(A\boldsymbol{v})_j = 2\sin(j\varphi) - \sin\big((j{+}1)\varphi\big) - \sin\big((j{-}1)\varphi\big) = (2 - 2\cos\varphi)\,\sin(j\varphi)$$

- **Imposing boundary condition（境界条件）:**
  - $v_0 = \sin 0 = 0$: automatic.
  - $v_{N+1} = \sin\big((N{+}1)\varphi\big) \overset{!}{=} 0 \implies \varphi_a = \dfrac{\pi a}{N+1}, \quad a = 1, \ldots, N.$

- **Result:**

$$v_j^{(a)} = \sin\frac{\pi a j}{N+1}, \qquad \lambda_a = 4\sin^2\frac{\pi a}{2(N+1)}.$$

Each **normal mode**（基準振動） oscillates rigidly, with frequency $\omega_a = \sqrt{\frac{k}{m}\lambda_a} = 2\sqrt{\frac{k}{m}} \sin \frac{\pi a}{2(N+1)}$:



mode $a = 1$ (slowest)　　mode $a = 2$　　mode $a = 3$

- The general motion is a *superposition*（重ね合わせ） of these modes — and writing a motion in this basis is exactly a discrete Fourier (sine) transformation:

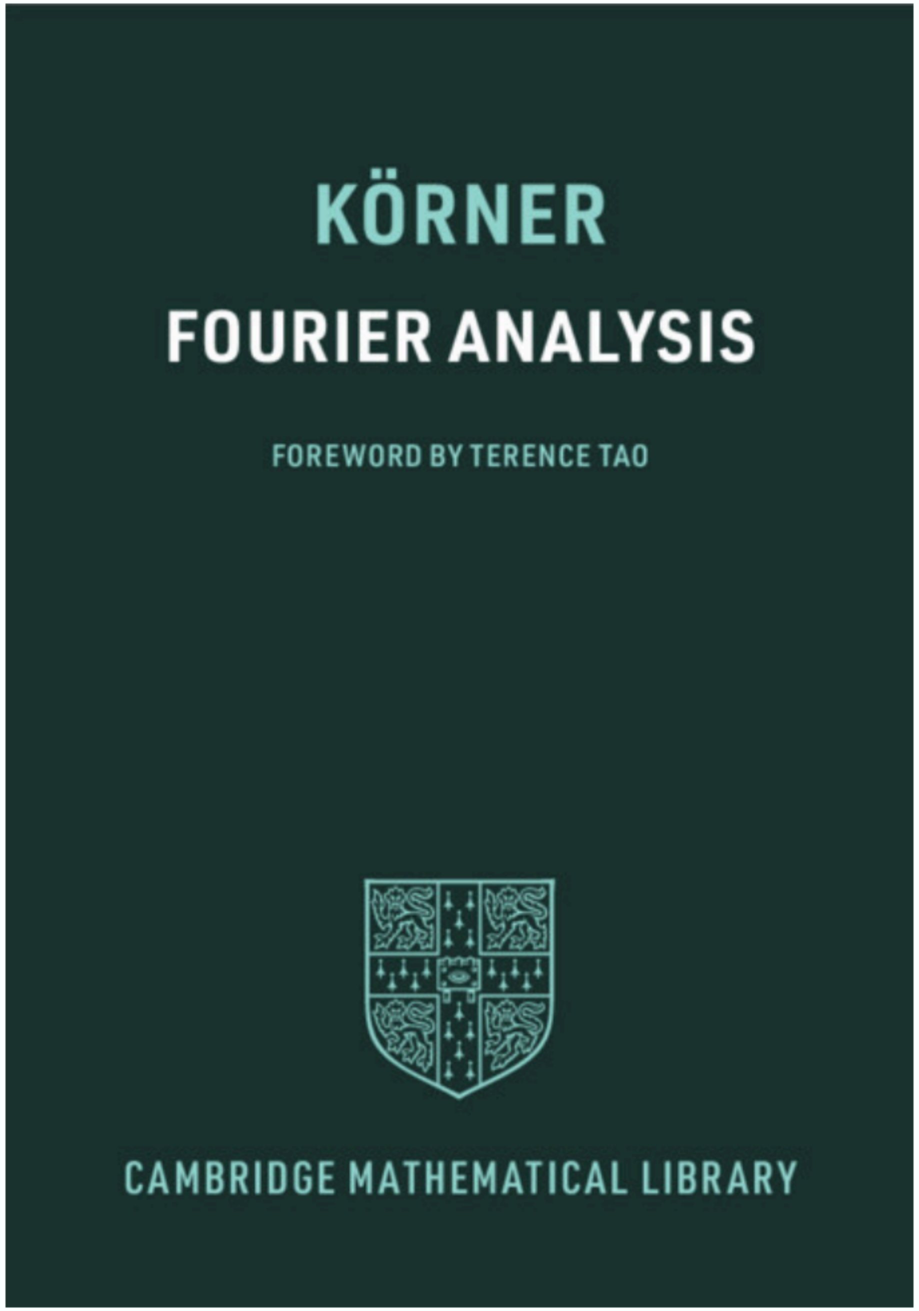
$$x_j(t) = \sum_{a=1}^{N} c_a \sin \frac{\pi a j}{N+1} \cos(\omega_a t + \varphi_a)$$

- For $N \to \infty$ the chain becomes a string: the equation becomes the **wave equation**（波動方程式）

$$\frac{\partial^2 x}{\partial t^2} = v^2 \frac{\partial^2 x}{\partial s^2}$$

and the sum over modes becomes the **Fourier series**（フーリエ級数）.


J. Boilly Del.
Geille Sculp.
Fourier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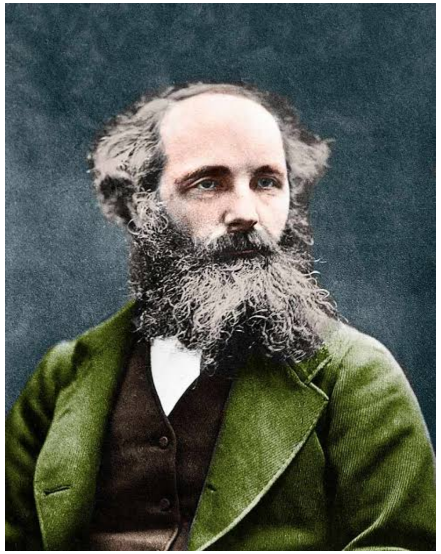
SECTION 2

Electromagnetism

電磁気学

All of electricity, magnetism, and optics in four lines:



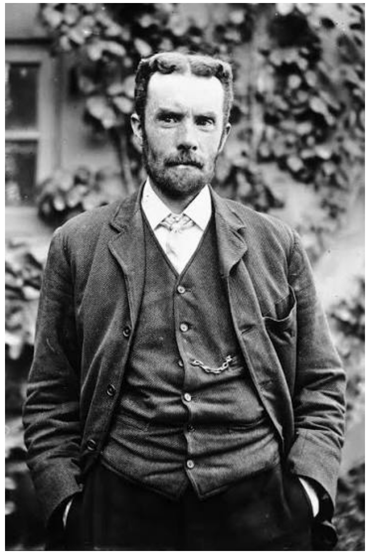
Maxwell

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \qquad \text{(Gauss)}$$

$$\nabla \cdot \vec{B} = 0 \qquad \text{(Gauss for magnetism)}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \qquad \text{(Faraday)}$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \qquad \text{(Ampère–Maxwell)}$$



Heaviside

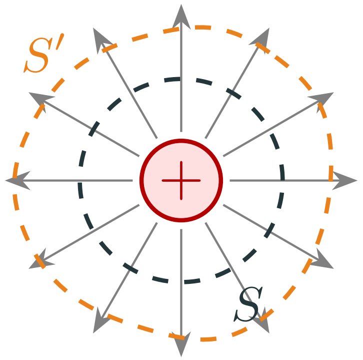
- In empty space they combine into a **wave equation** （波動方程式） with speed

$$\frac{1}{\sqrt{\mu_0 \varepsilon_0}} \approx 3 \times 10^8 \ \mathrm{m/s} \qquad \Longrightarrow \qquad \text{light is an electromagnetic wave.}$$

- The equations give this speed without mentioning any observer — a first hint of relativity (Section 4).

- **Stokes' theorem**（ストークスの定理） says: *integrating a derivative over a region gives the field on its boundary* ($\int_M d\omega = \int_{\partial M} \omega$).
- Applied to Gauss's law（ガウスの法則） $\nabla \cdot \vec{E} = \rho/\varepsilon_0$:

$$\int_V \nabla \cdot \vec{E}\, dV \;=\; \oint_{S=\partial V} \vec{E} \cdot d\vec{S} \;=\; \frac{Q_{\text{enc}}}{\varepsilon_0}$$

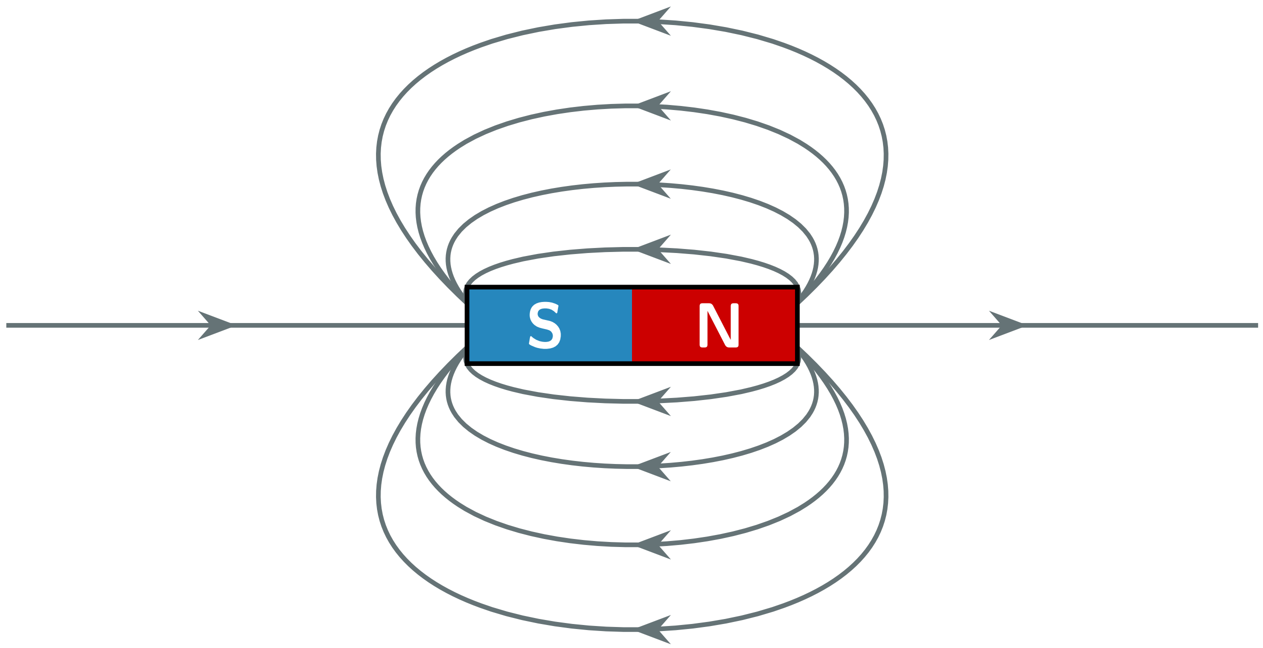


$$\vec{E} = \frac{q}{4\pi\varepsilon_0 r^2}\,\hat{r}$$

positive charge — source

- The shape of the volume V does not matter: deforming $S = \partial V$ into S' crosses no charge, so the flux（流束） is unchanged — only the charge inside counts.

How about the magnetic field $\vec{B}$?



Field lines never begin or end — every line that leaves N returns to S:

$$\nabla \cdot \vec{B} = 0$$

- $\nabla \cdot \vec{B} = 0$ says magnetic field lines never begin or end: there are no magnetic charges.
- For any vector field $\vec{A}$, the identity $\nabla \cdot (\nabla \times \vec{A}) = 0$ holds. So writing $\vec{B} = \nabla \times \vec{A}$ solves the equation *automatically* — and on $\mathbb{R}^3$ the converse is also true:

$$\vec{B} = \nabla \times \vec{A} \quad \Longleftrightarrow \quad \nabla \cdot \vec{B} = 0 \qquad \text{(on all of } \mathbb{R}^3\text{)}.$$

- $\vec{A}$ is the **vector potential**（ベクトルポテンシャル）; it is not unique: $\vec{A} \to \vec{A} + \nabla\chi$ gives the same $\vec{B}$ (**gauge freedom**（ゲージ自由度）).
- Is $\vec{A}$ real physics, or just bookkeeping?

- *Imagine* a magnetic charge: a radial $\vec{B}$ with total flux（流束） $\oint_{S^2} \vec{B}\cdot d\vec{S} = g \neq 0$ through a sphere.

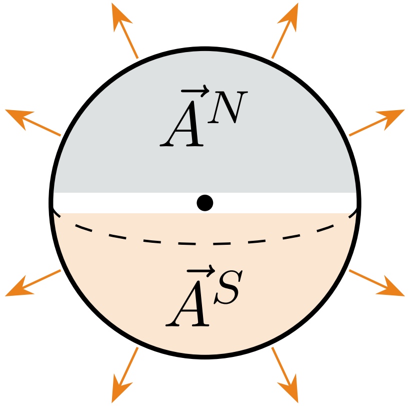
- But if $\vec{B} = \nabla \times \vec{A}$ everywhere, this flux must be zero: the hole at the origin changes the rules — topology（位相幾何学） enters.

- **Dirac's monopole（磁気単極子） (1931):** two potentials, one per hemisphere,

$$\vec{A}^{N/S} = \pm \frac{g}{4\pi r}\,\frac{1 \mp \cos\theta}{\sin\theta}\,\hat{\varphi},$$

  glued on the overlap by a *gauge transformation*（ゲージ変換） $\chi = g\varphi/2\pi$.

P. A. M. Dirac



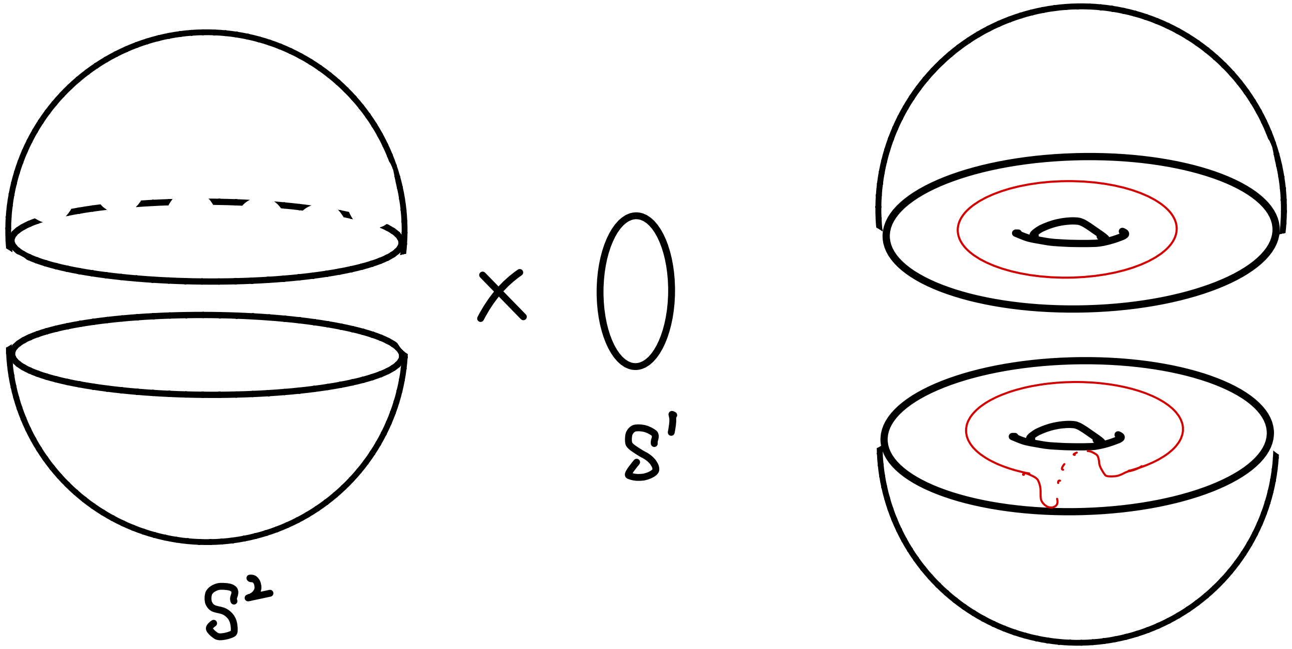
flux g through the sphere

Dirac's two glued potentials form a $U(1)$ **bundle over** S^2 — circles of phases glued with a *twist*.

- The simplest ($n = 1$) is the Hopf bundle（ホップ束） $S^1 \hookrightarrow S^3 \xrightarrow{\pi} S^2$: for $(z, w) \in \mathbb{C}^2$ with $|z|^2 + |w|^2 = 1$,

$$\pi(z, w) = \big(\, |z|^2 - |w|^2,\ 2\,\mathrm{Re}(z\bar{w}),\ 2\,\mathrm{Im}(z\bar{w}) \,\big) \ \in\ S^2.$$

- The total space is S^3, not $S^2 \times S^1$: the fibers are linked.



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- The total space is S^3, not $S^2 \times S^1$: the fibers are linked.
- The twist is measured by the **first Chern number** **（チャーン数）** (Chern 1946) of $F = dA$: $c_1 = \dfrac{1}{2\pi}\displaystyle\int_{S^2} F = n \in \mathbb{Z}$ (the monopole（磁気単極子） charge).

S.-S. Chern

Charge is quantized because the Chern number is an integer — and π will soon reappear as the *Bloch sphere*（ブロッホ球） of a spin.

Maxwell equations

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \qquad \text{(Gauss)}$$

$$\nabla \cdot \vec{B} = 0 \qquad \text{(Gauss for magnetism)}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \qquad \text{(Faraday)}$$

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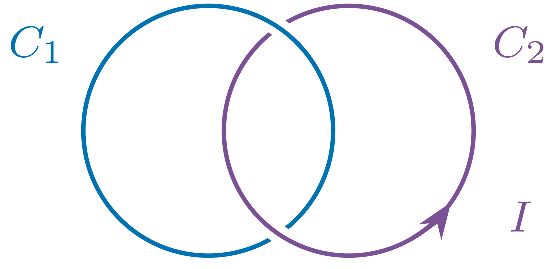
Gauss's linking integral

A current I around a loop C_2 creates a magnetic field (**Biot–Savart**（ビオ・サバールの法則） 1820):

$$\vec{B}(\vec{r}) = \frac{\mu_0 I}{4\pi} \oint_{C_2} \frac{d\vec{r}_2 \times (\vec{r} - \vec{r}_2)}{|\vec{r} - \vec{r}_2|^3}.$$



$\mathrm{Lk} = 1$

Ampère's law（アンペールの法則）: circulating $\vec{B}$ around a second loop C_1 counts how often the current threads it:

$$\oint_{C_1} \vec{B} \cdot d\vec{r}_1 = \mu_0 I \times \big(\#\ \text{times } C_2 \text{ threads } C_1\big).$$

$\mathrm{Lk} = 0$

Combining the two, the physics $(\mu_0 I)$ cancels — what remains is pure geometry:

$$\mathrm{Lk}(C_1, C_2) = \frac{1}{4\pi} \oint_{C_1} \oint_{C_2} \frac{(d\vec{r}_1 \times d\vec{r}_2) \cdot (\vec{r}_1 - \vec{r}_2)}{|\vec{r}_1 - \vec{r}_2|^3} \in \mathbb{Z} \qquad \text{(Gauss 1833).}$$

A double integral that always returns an integer: it cannot change under continuous deformation — the **linking number**（絡み数）, the first invariant（不変量） of knots and links.

[4.]

Von der *Geometria Situs*, die LEIBNITZ ahnte und in die nur einem Paar Geometern (EULER und VANDERMONDE) einen schwachen Blick zu thun vergönnt war, wissen und haben wir nach anderthalbhundert Jahren noch nicht viel mehr wie nichts.

Eine Hauptaufgabe aus dem *Grenzgebiet* der *Geometria Situs* und der *Geometria Magnitudinis* wird die sein, die Umschlingungen zweier geschlossener oder unendlicher Linien zu zählen.

Es seien die Coordinaten eines unbestimmten Punkts der ersten Linie x, y, z; der zweiten x', y', z' und

$$\iint \frac{(x'-x)(\mathrm{d}y\,\mathrm{d}z'-\mathrm{d}z\,\mathrm{d}y')+(y'-y)(\mathrm{d}z\,\mathrm{d}x'-\mathrm{d}x\,\mathrm{d}z')+(z-z')(\mathrm{d}x\,\mathrm{d}y'-\mathrm{d}y\,\mathrm{d}x')}{[(x'-x)^2+(y'-y)^2+(z'-z)^2]^{\frac{3}{2}}} = V$$

dann ist dies Integral durch beide Linien ausgedehnt

$$= 4m\pi$$

und m die Anzahl der Umschlingungen.

Der Werth ist gegenseitig, d. i. er bleibt derselbe, wenn beide Linien gegen einander umgetauscht werden. 1833. Jan. 22.

Gauss

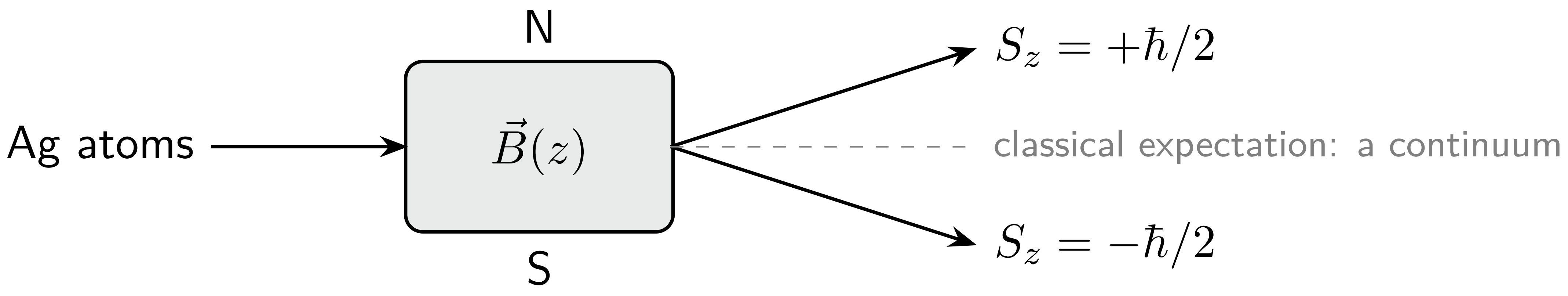
SECTION 3

Quantum Mechanics

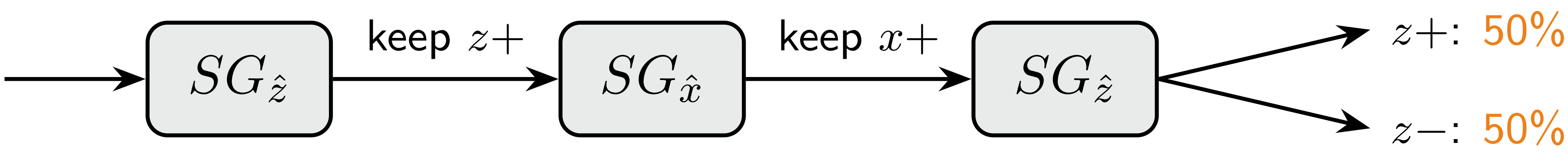
量子力学

Stern-Gerlach（シュテルン・ゲルラッハ） experiment (1922)

Send silver atoms through a *non-uniform* magnetic field.



- **Classical prediction:** a continuous smear.
- **Observed:** exactly two beams (up and down) — the spin is quantized: $S_z = \pm\hbar/2$.



- Select atoms with $S_z = +\hbar/2$. Measure S_x: result $\pm\hbar/2$, 50/50.
- Keep $S_x = +\hbar/2$ and measure S_z *again*: 50/50 — the earlier z-information is gone!
- No classical bookkeeping ("the atom has definite S_z *and* S_x") can reproduce this.

What mathematical structure produces such behavior? Linear algebra.

Postulates for spin-$\frac{1}{2}$:

- A state is a unit vector in $\mathbb{C}^2$, e.g. $|\uparrow\rangle = \binom{1}{0}$, $|\downarrow\rangle = \binom{0}{1}$.
- Spin observables（物理量）are built from the Hermitian（エルミート）**Pauli matrices**（パウリ行列）: $S_i = \frac{\hbar}{2}\sigma_i$,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- Their commutators（交換子） close among themselves:

$$[\sigma_i, \sigma_j] = 2i\,\epsilon_{ijk}\sigma_k, \qquad [S_i, S_j] = i\hbar\,\epsilon_{ijk}S_k.$$

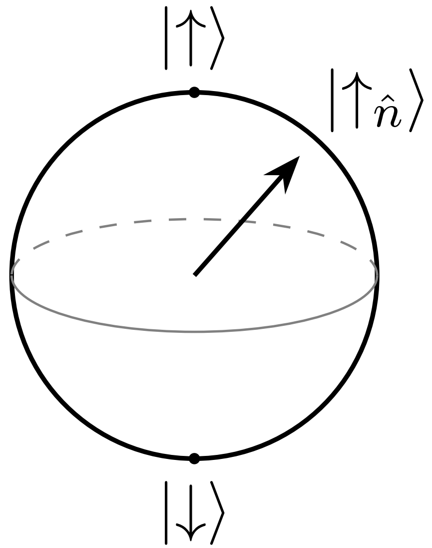
  These commutation relations define the $\mathfrak{su}(2)$ Lie algebra（リー代数）!

- Measurement outcomes = eigenvalues（固有値） (each σ_i has eigenvalues $\pm 1 \Rightarrow S_i = \pm\hbar/2$: two spots!).

The spin along a unit vector $\hat{n} = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta)$ is $\hat{n}\cdot\vec{\sigma}$, with eigenvector (固有ベクトル)

$$|{\uparrow_{\hat{n}}}\rangle = \begin{pmatrix} \cos\frac{\theta}{2} \\ e^{i\phi}\sin\frac{\theta}{2} \end{pmatrix}$$

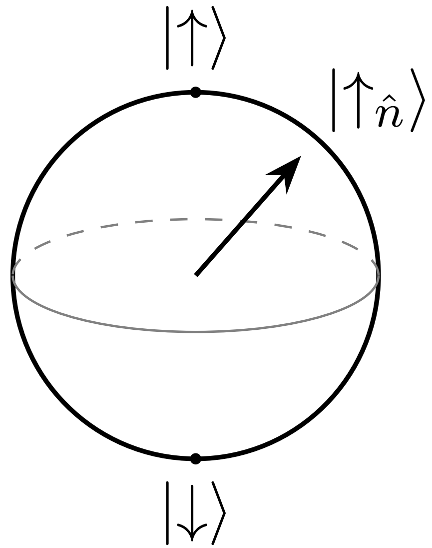
- Every spin state $\leftrightarrow$ a point of the sphere S^2 (the **Bloch sphere**（ブロッホ球）).
- Today, such a two-level system is called a qubit（量子ビット）— the unit of quantum information.



The spin along a unit vector $\hat{n} = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta)$ is $\hat{n}\cdot\vec{\sigma}$, with eigenvector (固有ベクトル)

$$|{\uparrow_{\hat{n}}}\rangle = \begin{pmatrix} \cos\frac{\theta}{2} \\ e^{i\phi}\sin\frac{\theta}{2} \end{pmatrix}$$

- A normalized spinor（スピノル） $(\alpha,\beta) \in \mathbb{C}^2$, $|\alpha|^2 + |\beta|^2 = 1$, is a point of S^3, and the overall phase $e^{i\chi}$ changes no physics. The map $(\alpha,\beta) \mapsto \hat{n}$ is exactly the Hopf map（ホップ写像） $S^1 \hookrightarrow S^3 \xrightarrow{\pi} S^2$ from Dirac's monopole (磁気単極子): **the Bloch sphere（ブロッホ球） is the base of the Hopf bundle（ホップ束）**.



- Probability $=$ $|$overlap with the eigenvector$|^2$; afterwards, the state *becomes* that eigenvector.
- The **Pauli matrix**（パウリ行列） σ_x has eigenvectors（固有ベクトル）

$$|{\uparrow_x}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \qquad |{\downarrow_x}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}.$$

- Start with $|{\uparrow}\rangle$ (selected $z+$). Probability of $x+$:

$$\left|\langle \uparrow_x \mid \uparrow\rangle\right|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2} \qquad \checkmark$$

- The state is now $|{\uparrow_x}\rangle$. Probability of $z+$:

$$\left|\langle \uparrow \mid \uparrow_x\rangle\right|^2 = \frac{1}{2} \qquad \checkmark$$

— the memory of $z+$ is erased.

- Non-commuting operators $[S_x, S_z] = -i\hbar S_y$ imply incompatible measurements. This is the seed of the uncertainty（不確定性） principle（不確定性原理）.

Uncertainty from non-commuting operators

- A quantum observable is a Hermitian（エルミート）matrix (operator). Its **uncertainty**（不確定性） in the state $|\psi\rangle$ is its spread:
$$(\Delta A)^2 = \langle A^2\rangle_\psi - \langle A\rangle_\psi^2.$$
- If A and B were both perfectly sharp in the same state, then
$$A|\psi\rangle = a|\psi\rangle, \quad B|\psi\rangle = b|\psi\rangle \quad \Longrightarrow \quad [A,B]|\psi\rangle = 0.$$
Therefore, when $[A,B]|\psi\rangle \neq 0$, the two observables（物理量） *cannot both be sharp*.
- Quantitatively, the **Robertson uncertainty relation**（ロバートソンの不確定性関係） is
$$\Delta A\,\Delta B \;\geq\; \frac{1}{2}\big|\langle [A,B]\rangle_\psi\big|.$$
- For spin, $[S_x, S_y] = i\hbar S_z$, so
$$\Delta S_x\,\Delta S_y \;\geq\; \frac{\hbar}{2}\,|\langle S_z\rangle|.$$

Position–momentum uncertainty（不確定性）

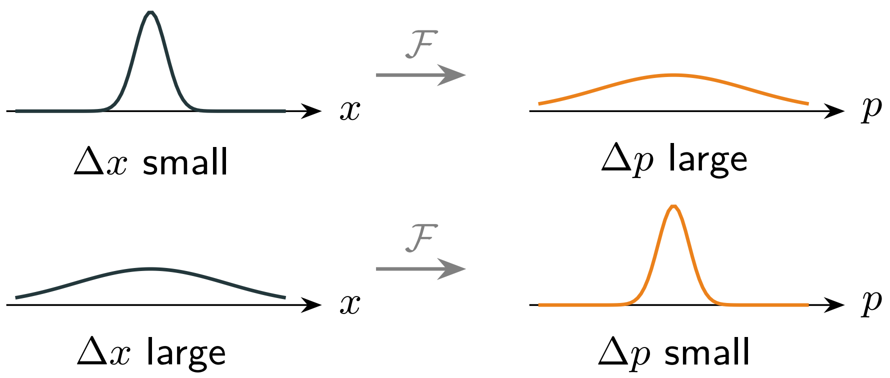
- Position and momentum *do not commute*: $[\hat{x}, \hat{p}] = i\hbar$. The Robertson relation then gives the uncertainty principle（不確定性原理）

$$\Delta x \, \Delta p \geq \frac{\hbar}{2} \qquad \text{— they cannot both be sharp.}$$

- The reason is Fourier: the momentum wave function（波動関数） is the Fourier transform（フーリエ変換） of $\psi(x)$ — the same transformation that diagonalized our springs:

$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int \psi(x) \, e^{-ipx/\hbar} \, dx,$$

and a function and its Fourier transform *cannot both be narrow*:



The Schrödinger equation and time–energy uncertainty（不確定性）

- In the position representation the momentum operator becomes $\hat{p} = -i\hbar \dfrac{d}{dx}$ — after a Fourier transformation（フーリエ変換）, $-i\hbar\, d/dx$ is simply multiplication by p.
- The energy then becomes the **Hamiltonian（ハミルトニアン） operator**（ハミルトニアン演算子）

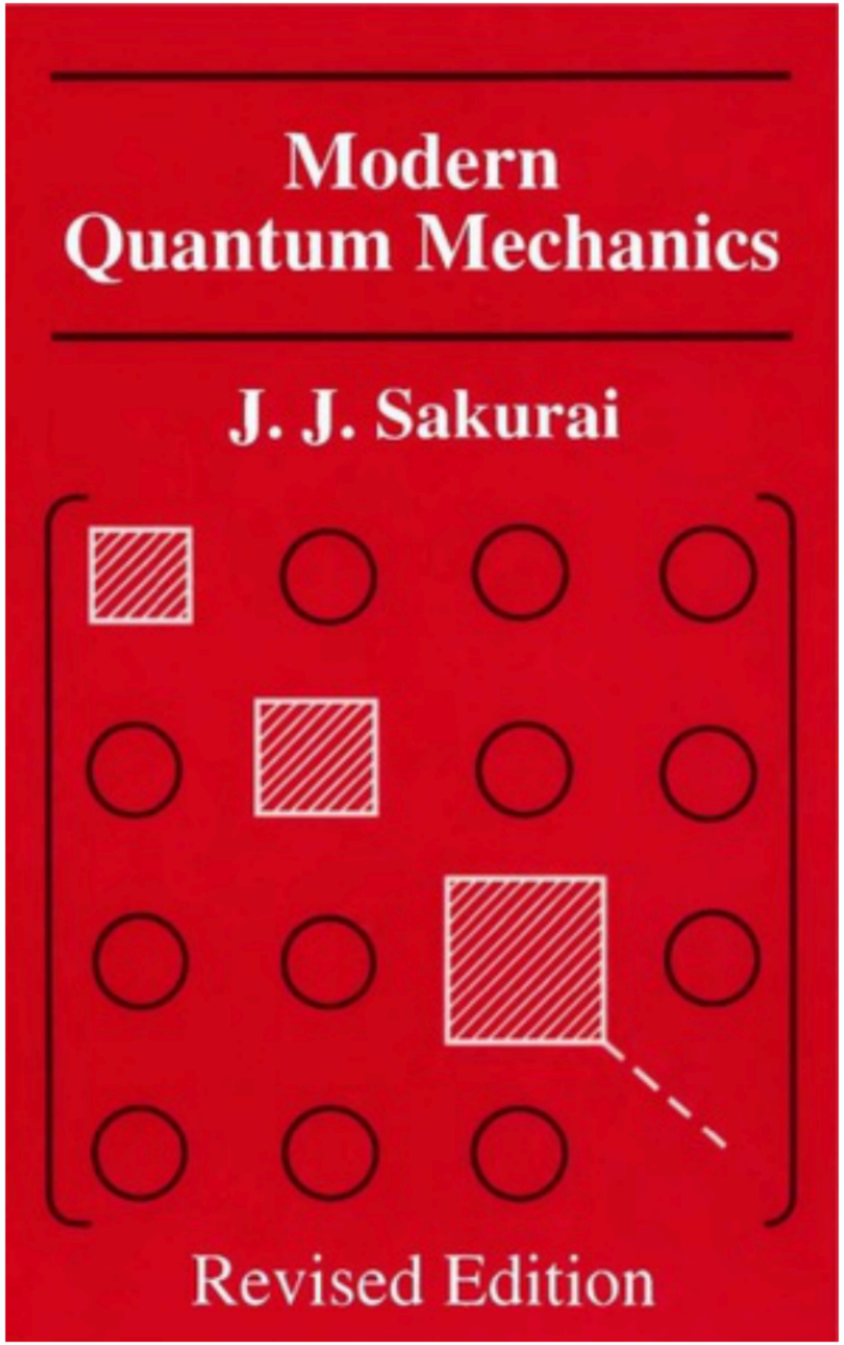
$$\hat{H} = \frac{\hat{p}^{\,2}}{2m} + V(\hat{x}) = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x),$$

  and the dynamics of ψ is the Schrödinger equation（シュレーディンガー方程式） $i\hbar \dfrac{\partial\psi}{\partial t} = \hat{H}\psi.$

- **Time–energy:** time is a *parameter*, not an operator. From the Schrödinger equation one derives (Mandelstam–Tamm)

$$\Delta H\, \tau_A \;\geq\; \frac{\hbar}{2}, \qquad \tau_A = \frac{\Delta A}{\big|d\langle A\rangle/dt\big|} \;=\; \text{time for } \langle A\rangle \text{ to shift by one } \Delta A.$$

  A state with a small energy spread can only change *slowly*.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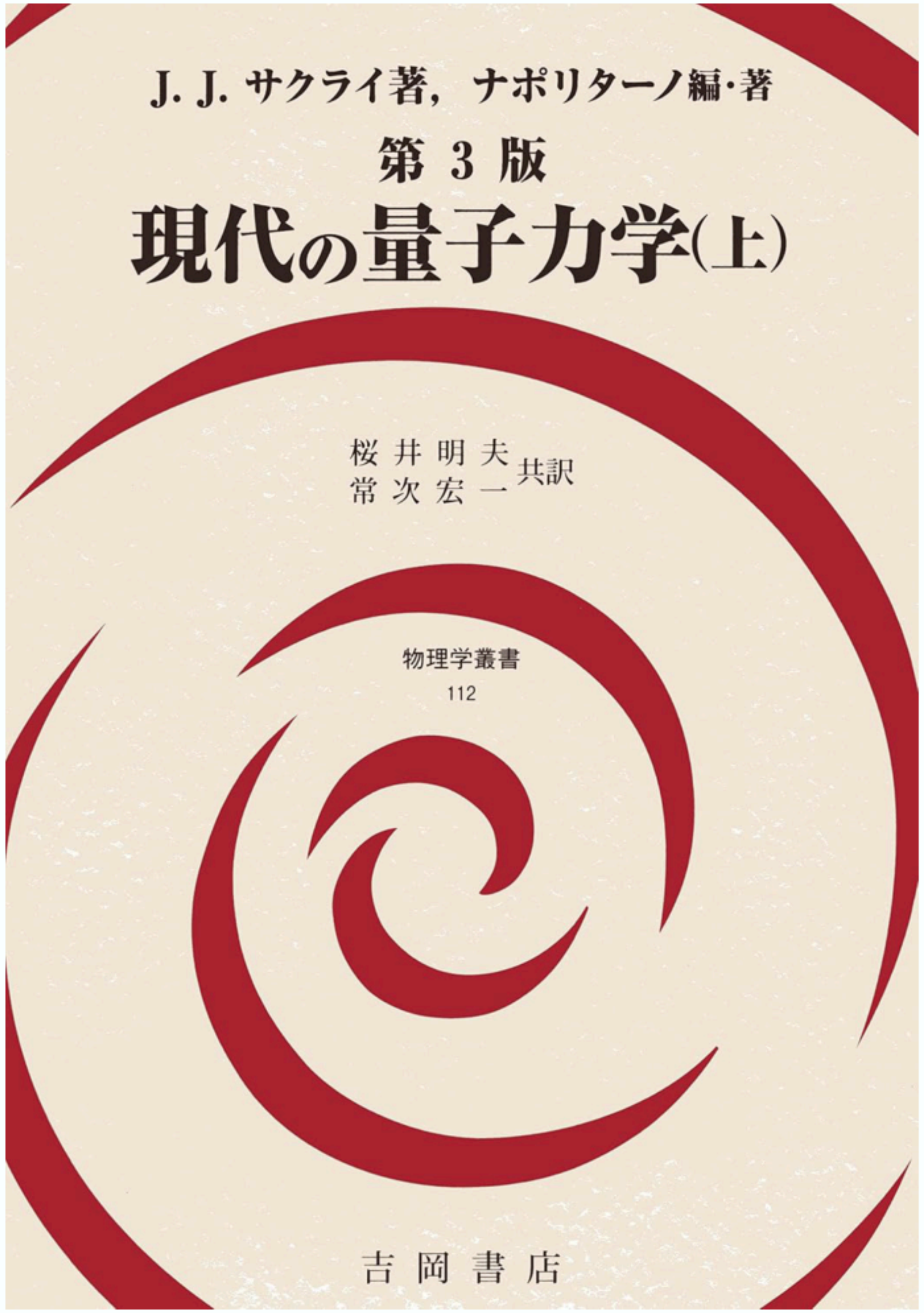
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SECTION 4

Special Relativity

特殊相対性理論

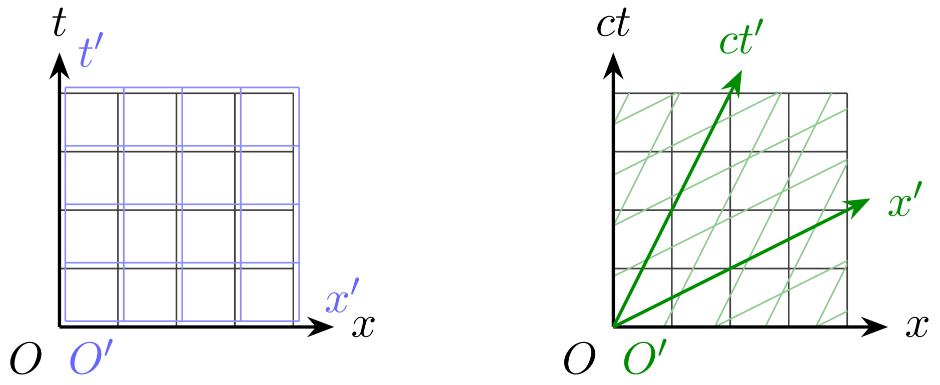
From the Galilei transformation（ガリレイ変換）to the Lorentz transformation（ローレンツ変換）

Einstein's postulates（仮説） (1905)

1. The laws of physics are the same in every inertial frame（慣性系）.
2. The speed of light c is the same for *all* inertial observers.

- **Galilei:** time and distances are *separately* invariant（不変量）: $t' = t,\ x_2' - x_1' = x_2 - x_1$ — the two grids coincide.
- **Lorentz:** light-speed invariance forces a different invariant: only the mixture

$$(x_2' - x_1')^2 - c^2(t_2' - t_1')^2 = (x_2 - x_1)^2 - c^2(t_2 - t_1)^2$$

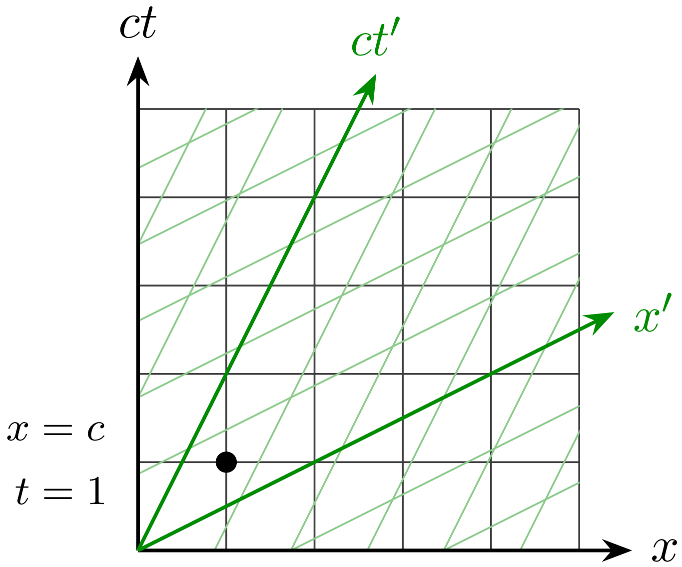


- Both transformations are *linear* — 2×2 matrices:

$$\underbrace{\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}}_{\textbf{rotation:}\text{ preserves } x^2+y^2} \qquad \underbrace{\begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} \cosh\eta & -\sinh\eta \\ -\sinh\eta & \cosh\eta \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}}_{\textbf{boost:}\text{ preserves } c^2t^2-x^2}$$

- A boost（ブースト） is a hyperbolic rotation（双曲回転） of spacetime（時空）; the "angle" η is fixed by the velocity: $\tanh\eta = v/c$ and $\cosh\eta = 1/\sqrt{1-(\frac{v}{c})^2}$.

- Example:

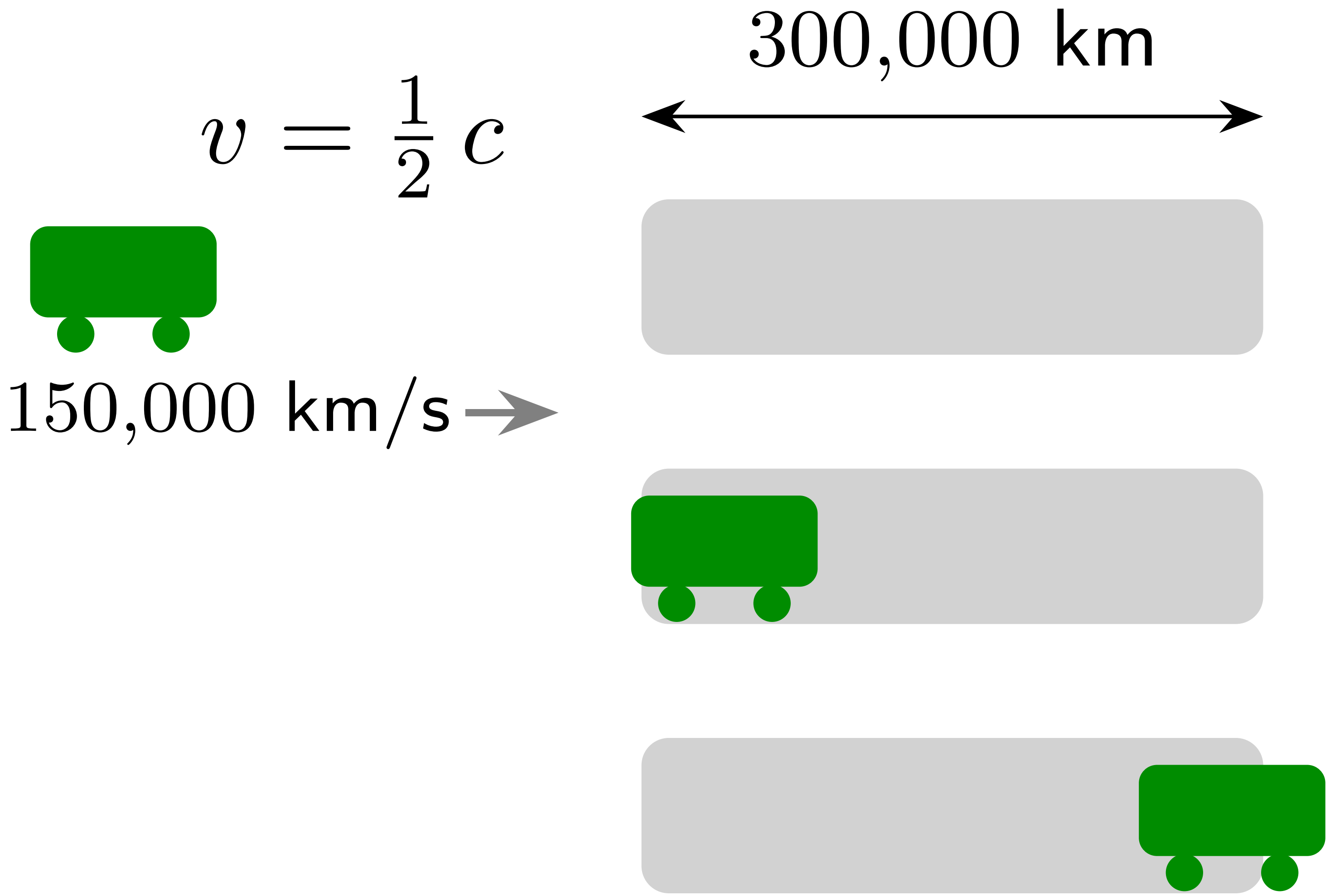


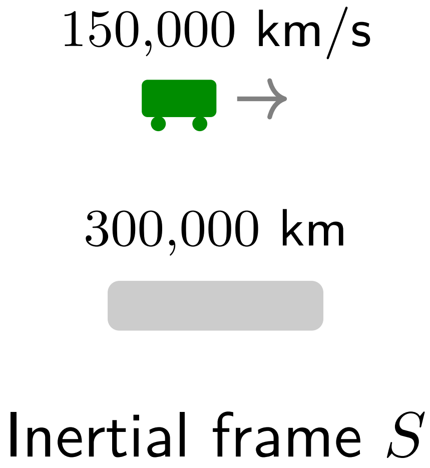
$$v = \frac{1}{2}c$$

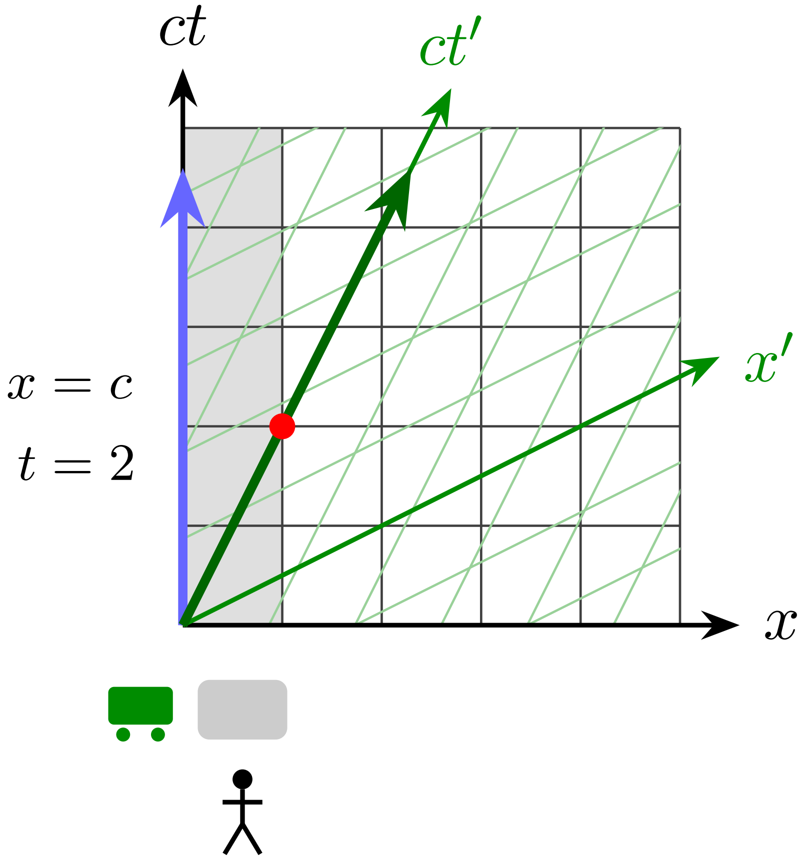
$$x' = \frac{1}{\sqrt{1-(\frac{v}{c})^2}}(x - vt) = \frac{1}{\sqrt{3}}c$$

$$t' = \frac{1}{\sqrt{1-(\frac{v}{c})^2}}\left(-\frac{v}{c^2}x + t\right) = \frac{1}{\sqrt{3}}$$

Inertial frame S Inertial frame S'


300,000 km
$v = \frac{1}{2}\,c$
150,000 km/s





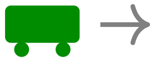
Inertial frame S'

$$t' = \frac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} \left(-\frac{v}{c^2}\, x + t\right)$$

$$= 1.73$$

The train passes through the tunnel in 1.73 s.

150,000 km/s

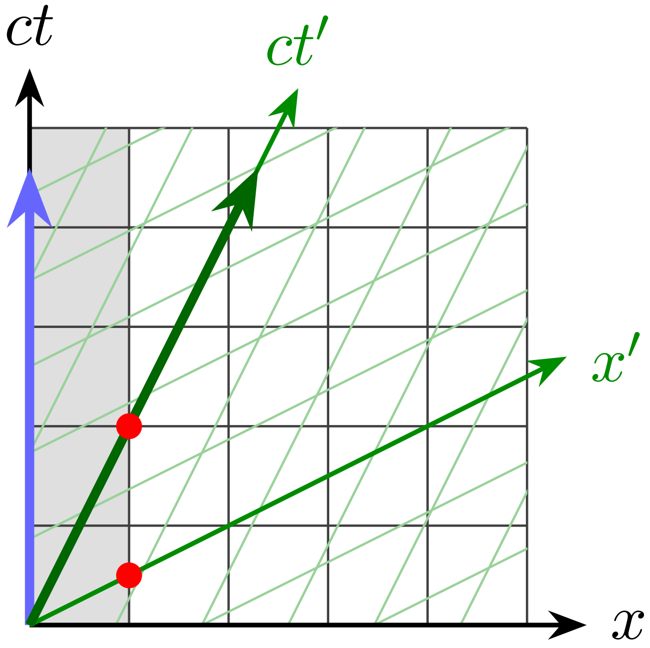


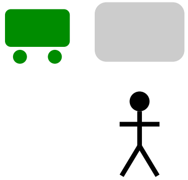
300,000 km

Inertial frame S

$x = c,\ t = 2$

$t' = 1.73$





$x = c,\ t = \frac{1}{2}$

Inertial frame S'

$$x' = \frac{1}{\sqrt{1-\left(\frac{v}{c}\right)^2}}(x - vt)$$

$$= \frac{\sqrt{3}}{2} \times 300{,}000 \text{ km}$$

$$\approx 260{,}000 \text{ km}$$

For the train, the tunnel is 260,000 km long.

In $3+1$ dimensions a Lorentz transformation（ローレンツ変換） is a *real* 4×4 matrix — yet *complex* 2×2 matrices tell the whole story.

- $\Lambda \in SO^{+}(1,3)$: the 4×4 matrices preserving the **interval**（世界間隔）, $\Lambda^{\mathsf{T}} \eta \, \Lambda = \eta$. Parameters: 6 = 3 rotations + 3 boosts.

- **The trick:** package a spacetime（時空）point into a **Hermitian**（エルミート）2×2 matrix (the **Pauli matrices**（パウリ行列） again!):

$$X = ct\,\mathbf{1} + \vec{x} \cdot \vec{\sigma} = \begin{pmatrix} ct + z & x - iy \\ x + iy & ct - z \end{pmatrix}, \qquad \det X = c^2 t^2 - x^2 - y^2 - z^2.$$

- For $M \in SL(2, \mathbb{C})$, the map $X \mapsto M X M^{\dagger}$ keeps X **Hermitian** and preserves $\det X \Rightarrow$ a Lorentz transformation. Also 6 parameters.

- *Every* one arises this way, with M, $-M$ giving the same Λ — *two-to-one*:

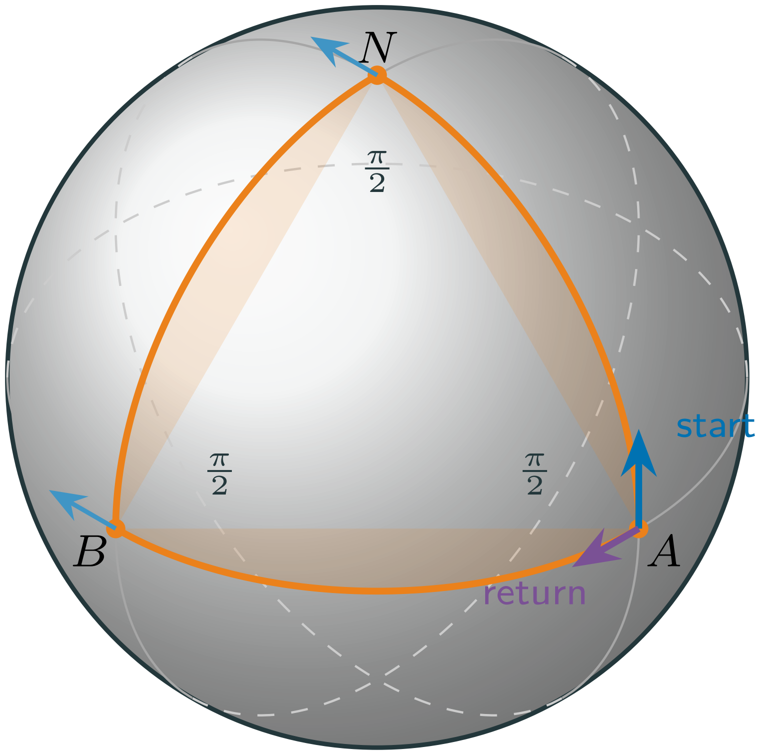
$$SL(2, \mathbb{C}) \,/ \{\pm 1\} \;\cong\; SO^{+}(1,3).$$

SECTION 5

General Relativity

一般相対性理論

Curved space



- The round sphere of unit radius has

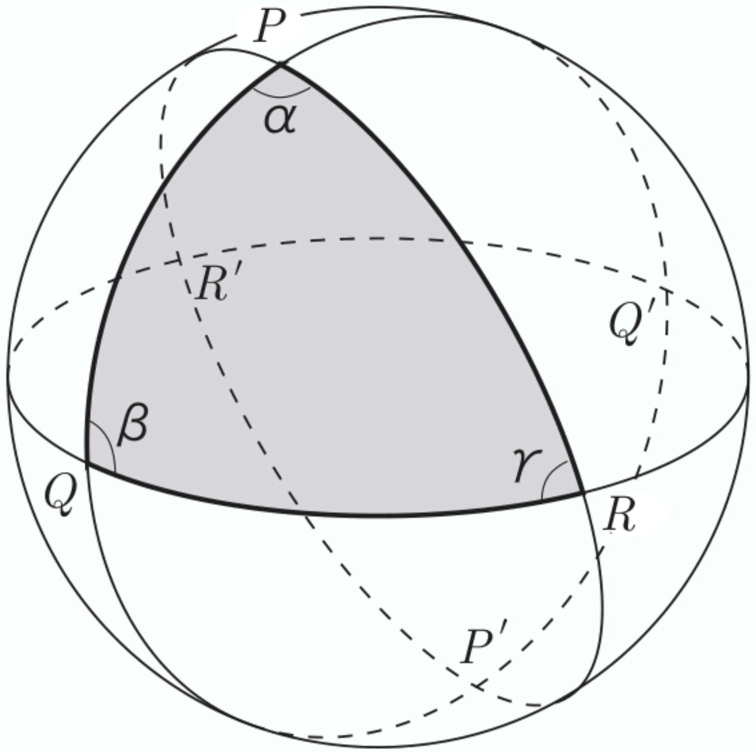
$$ds^2 = d\theta^2 + \sin^2\theta\, d\phi^2, \qquad K = 1.$$

- Three orthogonal planes cut out an octant: its sides are **great-circle geodesics** (測地線) and it is $\frac{1}{8}$ of the sphere,

$$\mathrm{Area}(\triangle) = \tfrac{1}{8} \cdot 4\pi = \frac{\pi}{2}.$$

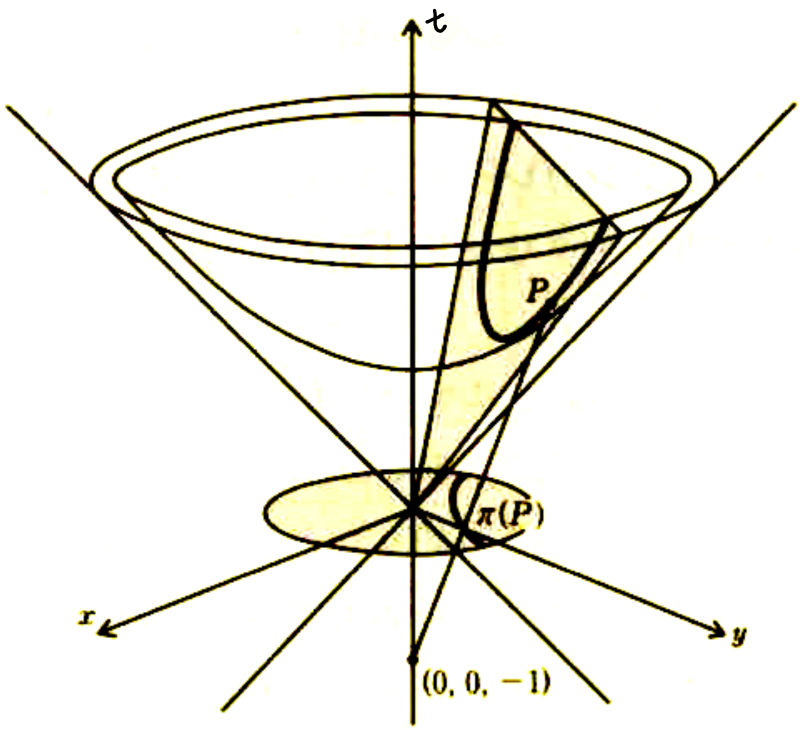
- **Parallel transport** around $A \to N \to B \to A$ never turns the vector *relative to the surface*, yet it comes back rotated:

$$\Delta\psi = \int_\triangle K\, dA = \mathrm{Area}(\triangle) = \frac{\pi}{2}.$$



$$\text{Area}(\triangle) = \alpha + \beta + \gamma - \pi$$

Hyperboloid and Poincare disk



- In 3D Minkowski space（ミンコフスキー空間）, the surface

  $$t^2-x^2-y^2=1, \quad (t=\cosh\rho,\ x+iy=\sinh\rho\, e^{i\phi})$$

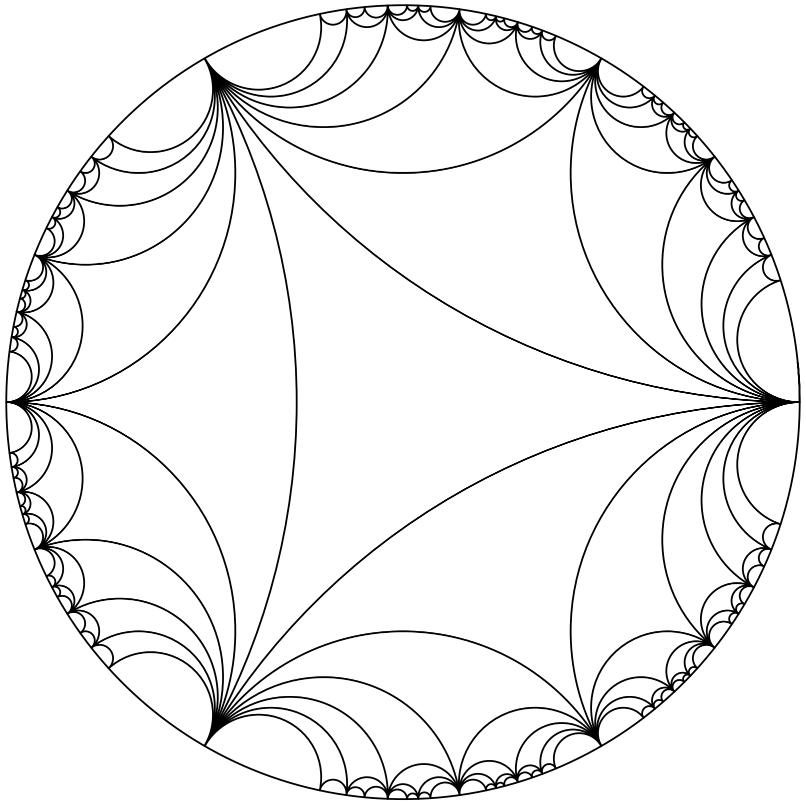
  inherits the metric（計量） $d\ell^2 = d\rho^2 + \sinh^2\rho\, d\phi^2$, of constant curvature（曲率） $K=-1$.

- Projecting from $(X^0,x,y) = (-1,0,0)$ maps the sheet onto the unit disk $|z|<1$ with $|z|=\tanh(\rho/2)$, giving the **Poincaré metric**（ポアンカレ計量）

  $$d\ell^2 = \frac{4\,|dz|^2}{\left(1-|z|^2\right)^2}.$$

- **Geodesics** are diameters and circular arcs *orthogonal* to the boundary.

Hyperboloid and Poincare disk



- In 3D Minkowski space（ミンコフスキー空間）, the surface

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inherits the metric（計量） $d\ell^2 = d\rho^2 + \sinh^2\rho\, d\phi^2$, of constant curvature（曲率） $K = -1$.

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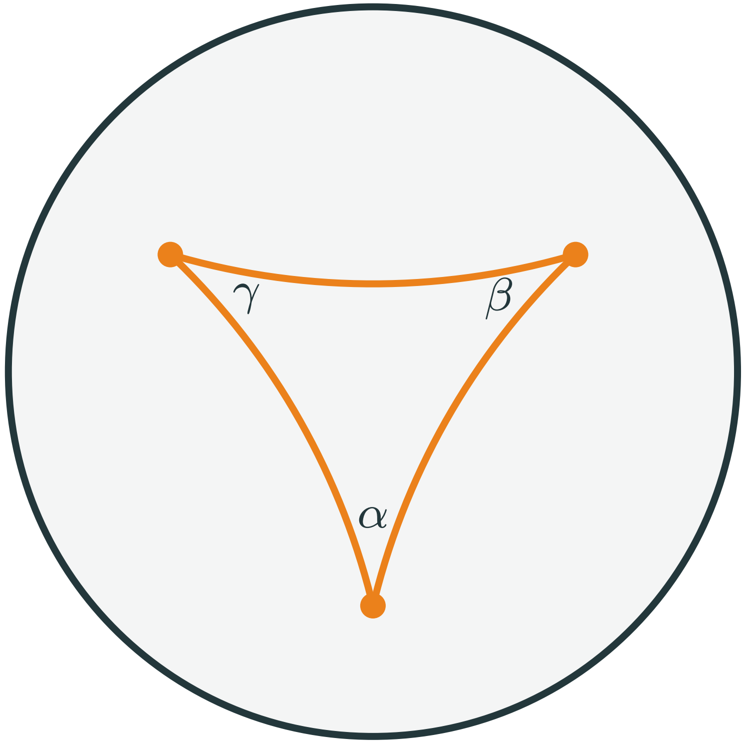
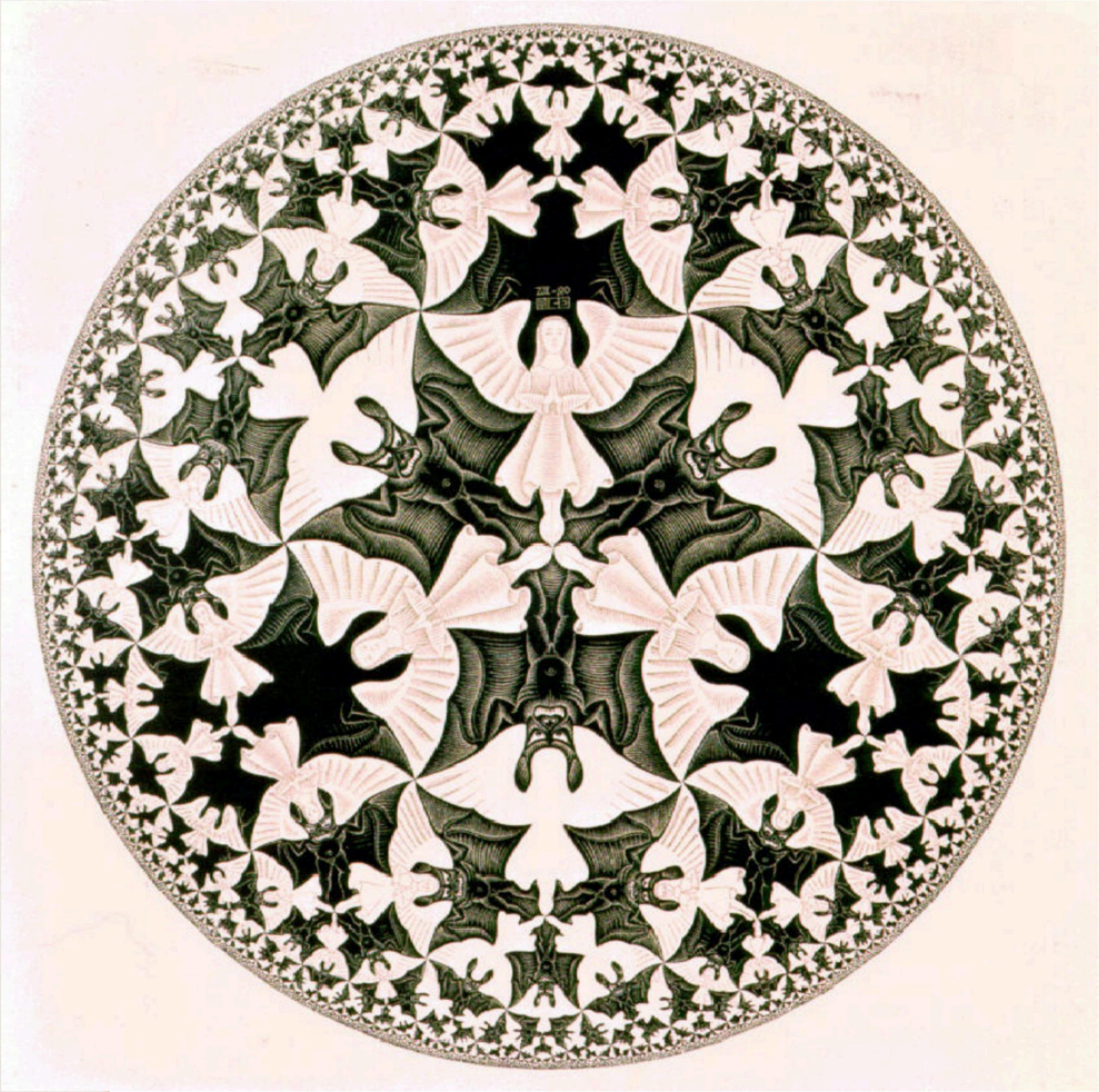
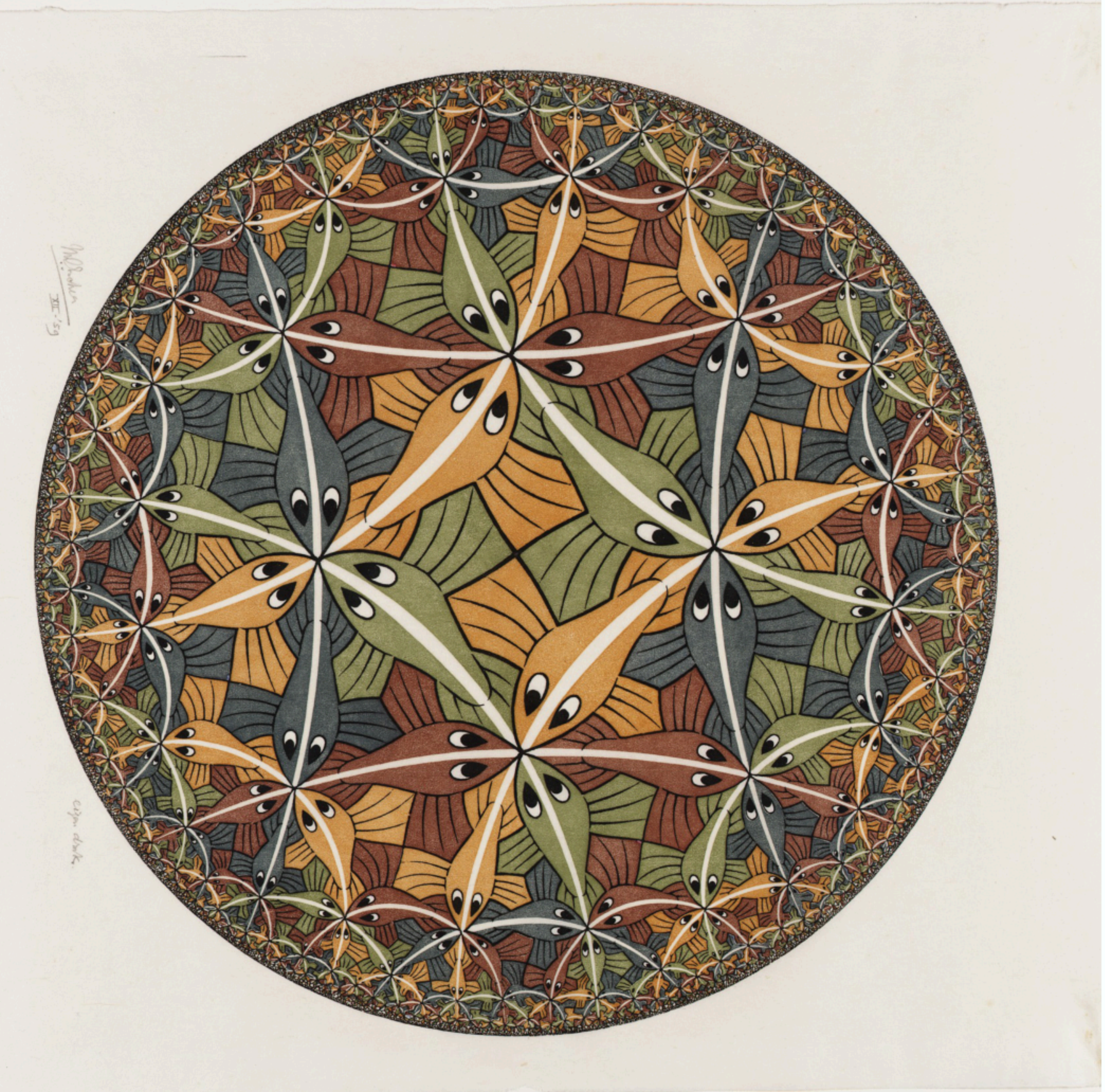
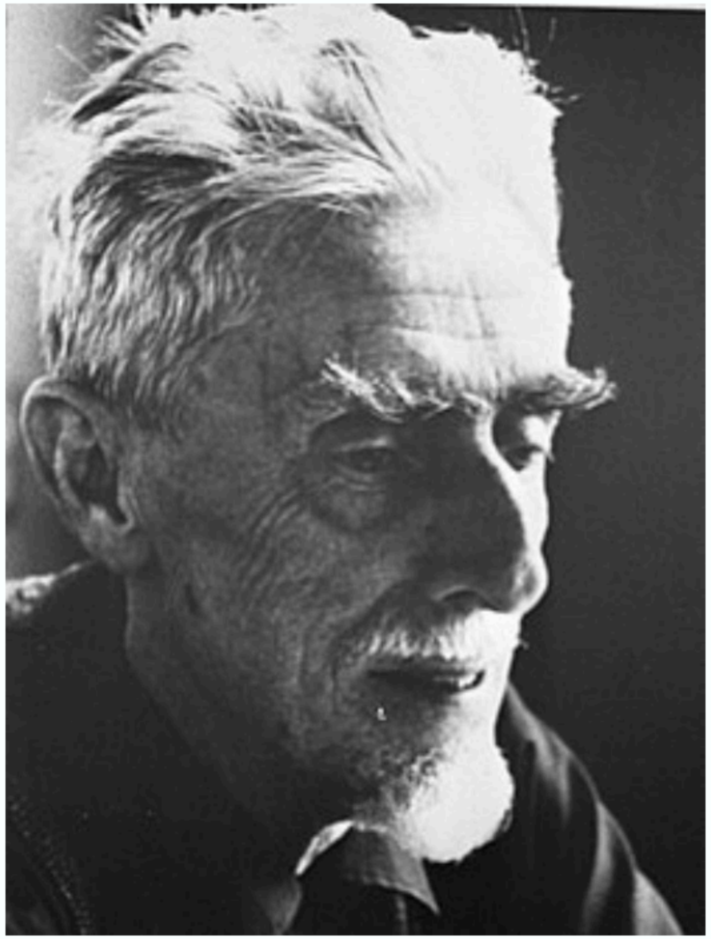


Poincaré disk（ポアンカレ円板）

$$\mathrm{Area}(\triangle) = \pi - (\alpha + \beta + \gamma)$$







M.C. Escher

Riemannian geometry: one field $g_{\mu\nu}$ rules everything

- A **Riemannian manifold（多様体）**（リーマン多様体） is a space that locally looks like $\mathbb{R}^n$, with a **metric**（計量） — a position-dependent ruler:

$$ds^2 = g_{\mu\nu}(x)\,dx^\mu dx^\nu \qquad \text{(sphere: } g_{\theta\theta} = R^2,\ g_{\phi\phi} = R^2 \sin^2\theta).$$

- **Geodesics** — the straightest paths, extremizing $\int ds$:

$$\frac{d^2x^\rho}{d\tau^2} + \Gamma^\rho{}_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0, \qquad \Gamma^\rho{}_{\mu\nu} = \tfrac{1}{2}\,g^{\rho\lambda}\big(\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu}\big),$$

  with the **Christoffel symbols**（クリストッフェル記号） Γ built from the metric alone.

- **Riemann curvature（曲率）**（リーマン曲率） — how parallel transport（平行移動） around a small loop rotates a vector ($\Delta\psi$ on the sphere; on a surface it is Gauss's K, and flat $\Leftrightarrow R^\rho{}_{\sigma\mu\nu} = 0$):

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho{}_{\nu\sigma} - \partial_\nu\Gamma^\rho{}_{\mu\sigma} + \Gamma^\rho{}_{\mu\lambda}\Gamma^\lambda{}_{\nu\sigma} - \Gamma^\rho{}_{\nu\lambda}\Gamma^\lambda{}_{\mu\sigma}, \qquad R_{\mu\nu} = R^\rho{}_{\mu\rho\nu}, \quad R = g^{\mu\nu}R_{\mu\nu}.$$

General covariance + equivalence principle (等価原理) $\Rightarrow$ Einstein's equation

equivalence principle

in free fall gravity vanishes: $g_{\mu\nu}(p) = \eta_{\mu\nu}$, ${\Gamma^\rho}_{\mu\nu}(p) = 0$

only tidal gravity ($\partial\Gamma \sim$ curvature) survives

general covariance

laws take the same form in every coordinate system

$\Rightarrow$ they must be tensor equations

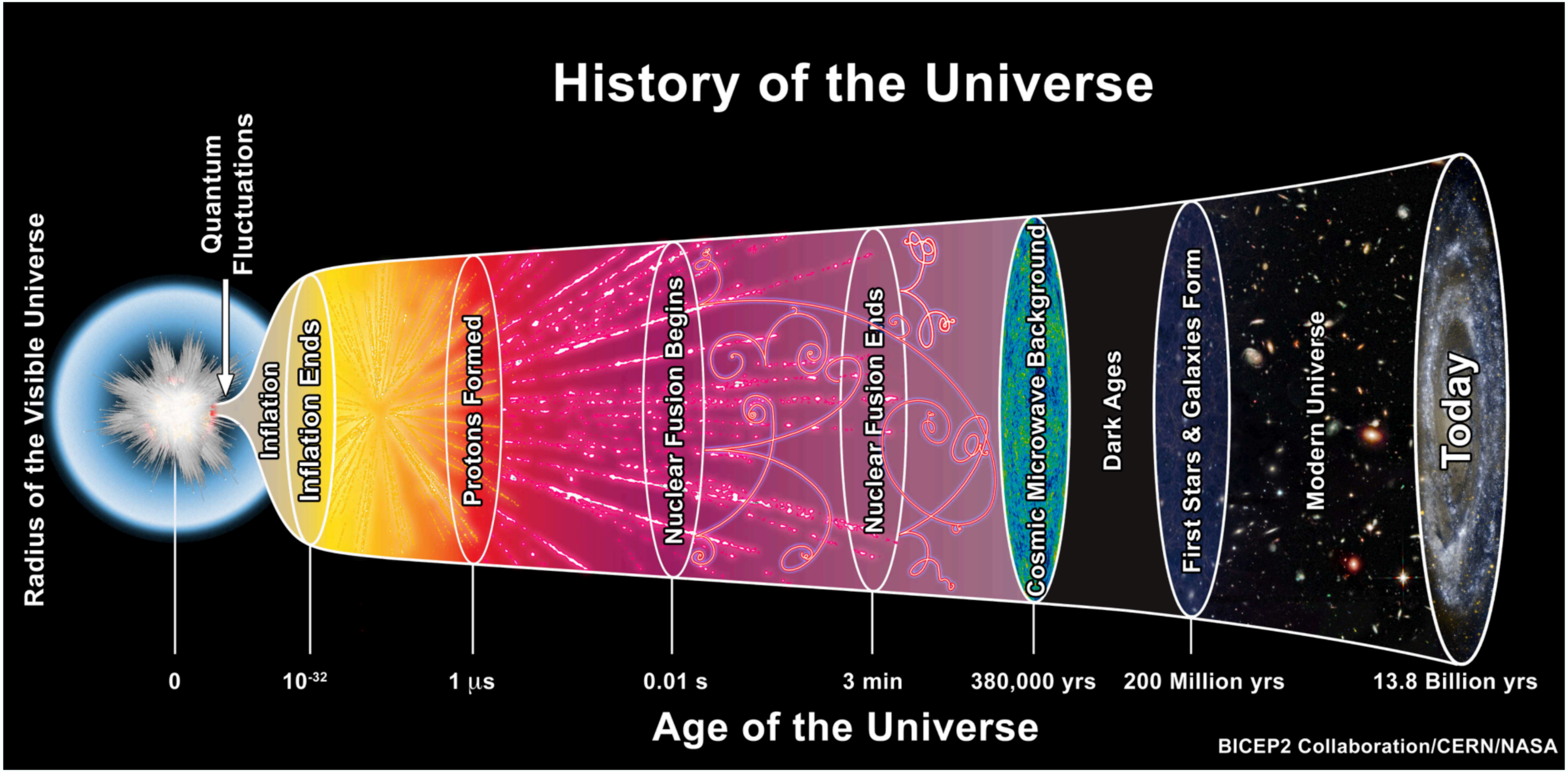
gravity is geometry: a tensor equation for $g_{\mu\nu}$, sourced by energy–momentum $T_{\mu\nu}$

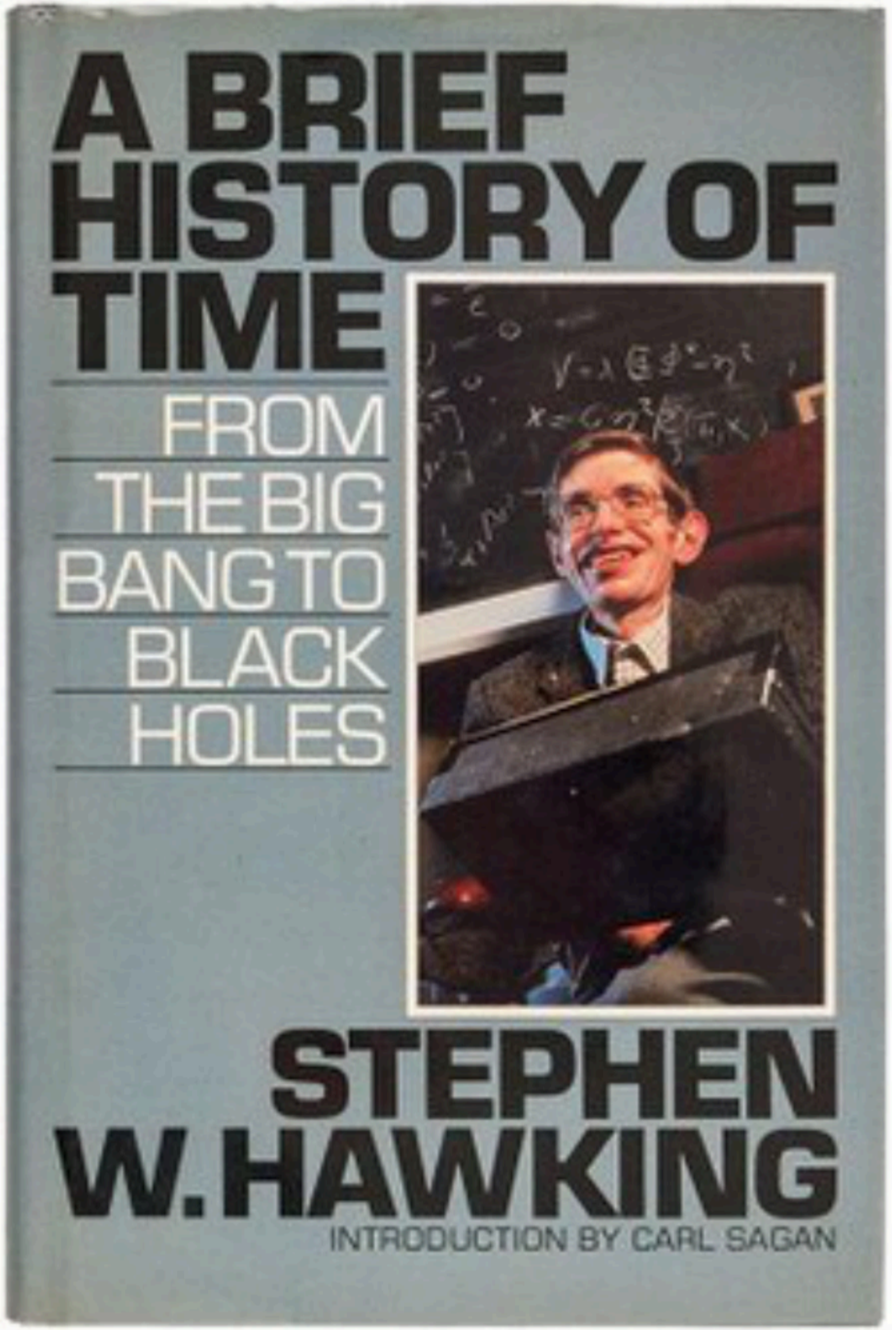
Conservation $\nabla^\mu T_{\mu\nu} = 0$ demands a *divergence-free* tensor（テンソル） built from $g_{\mu\nu}$ with at most two derivatives — and the Bianchi identity（ビアンキ恒等式） provides exactly one (uniquely so: Lovelock 1971):

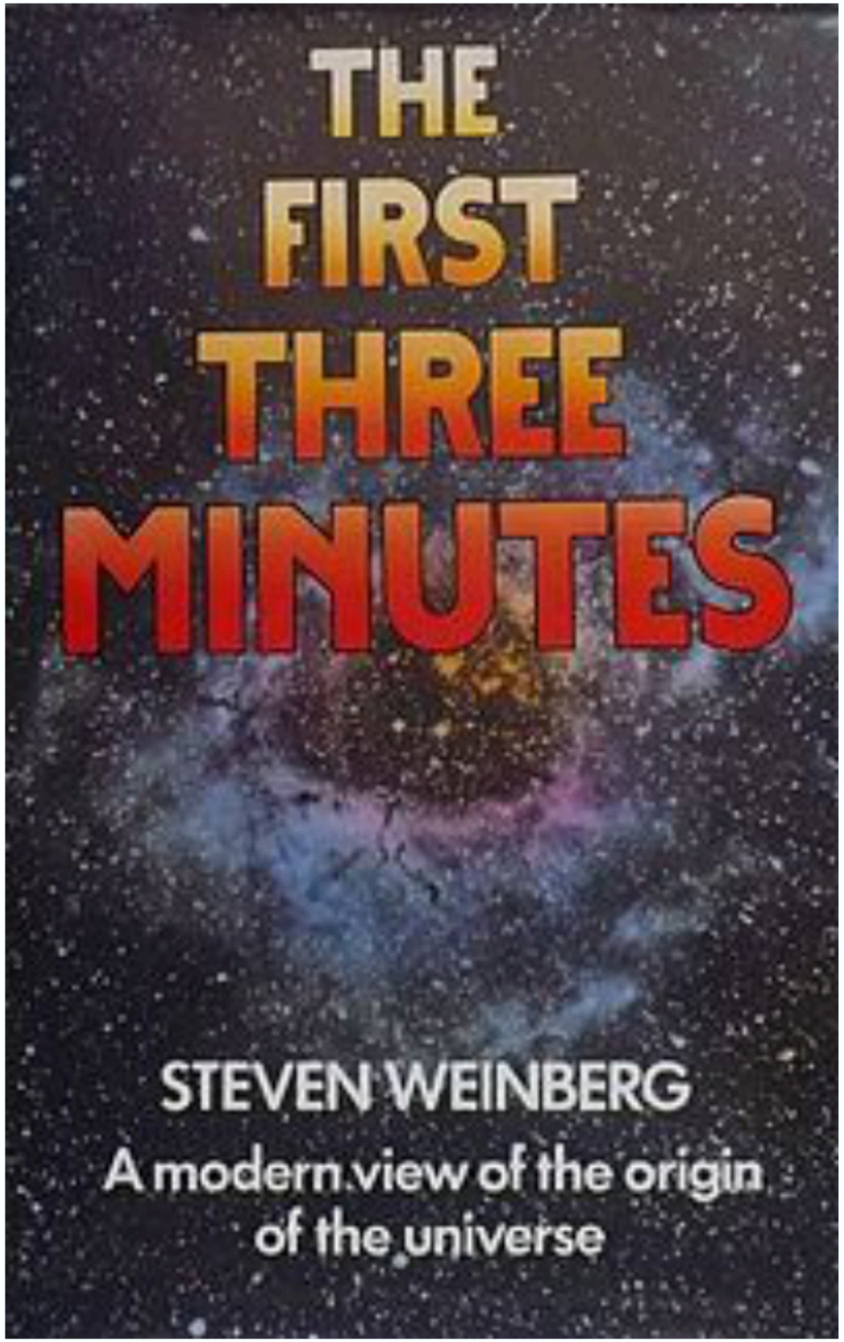
$$\nabla^\mu\left(R_{\mu\nu} - \tfrac{1}{2}\,R\,g_{\mu\nu}\right) = 0.$$

Recovering Newton's $\nabla^2\Phi = 4\pi G\rho$ for weak fields fixes the constant:

$$R_{\mu\nu} - \tfrac{1}{2}\,R\,g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}\,T_{\mu\nu} \qquad \text{(Einstein 1915).}$$


History of the Universe
Radius of the Visible Universe
Quantum Fluctuations
Inflation
Inflation Ends
Protons Formed
Nuclear Fusion Begins
Nuclear Fusion Ends
Cosmic Microwave Background
Dark Ages
First Stars & Galaxies Form
Modern Universe
Today
0
10^{-32}
1 μs
0.01 s
3 min
380,000 yrs
200 Million yrs
13.8 Billion yrs
Age of the Universe
BICEP2 Collaboration/CERN/NASA

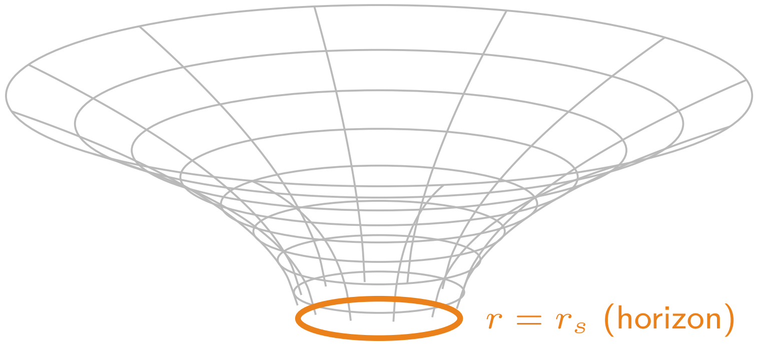

A BRIEF HISTORY OF TIME
FROM THE BIG BANG TO BLACK HOLES
STEPHEN W. HAWKING
INTRODUCTION BY CARL SAGAN


THE FIRST THREE MINUTES
STEVEN WEINBERG
A modern view of the origin of the universe

The Schwarzschild（シュワルツシルト）black hole（ブラックホール）and its causal structure（因果構造）

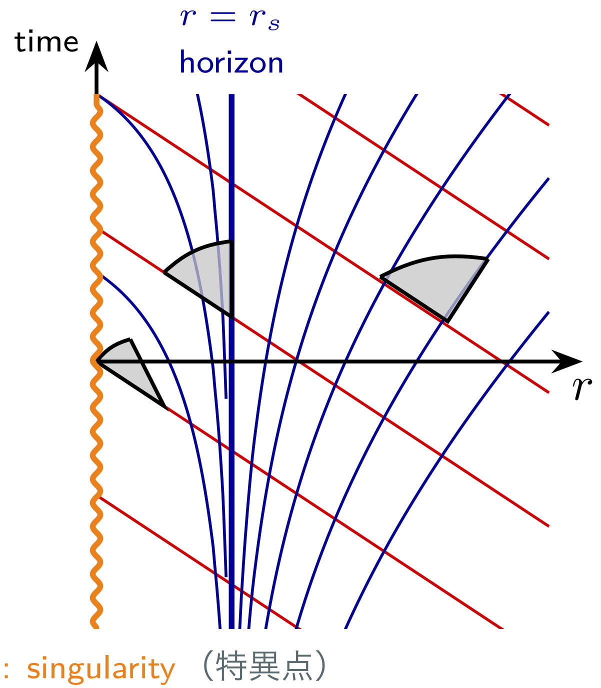
The vacuum solution（真空解） outside a spherical mass M (Schwarzschild 1916), with the Schwarzschild radius（シュワルツシルト半径） $r_s = 2GM/c^2$:

$$ds^2 = -\left(1 - \frac{r_s}{r}\right)c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2\left(d\theta^2 + \sin^2\theta\, d\phi^2\right).$$



space is stretched: the throat is measured distance;
clocks run slow（時間の遅れ） near the mass:

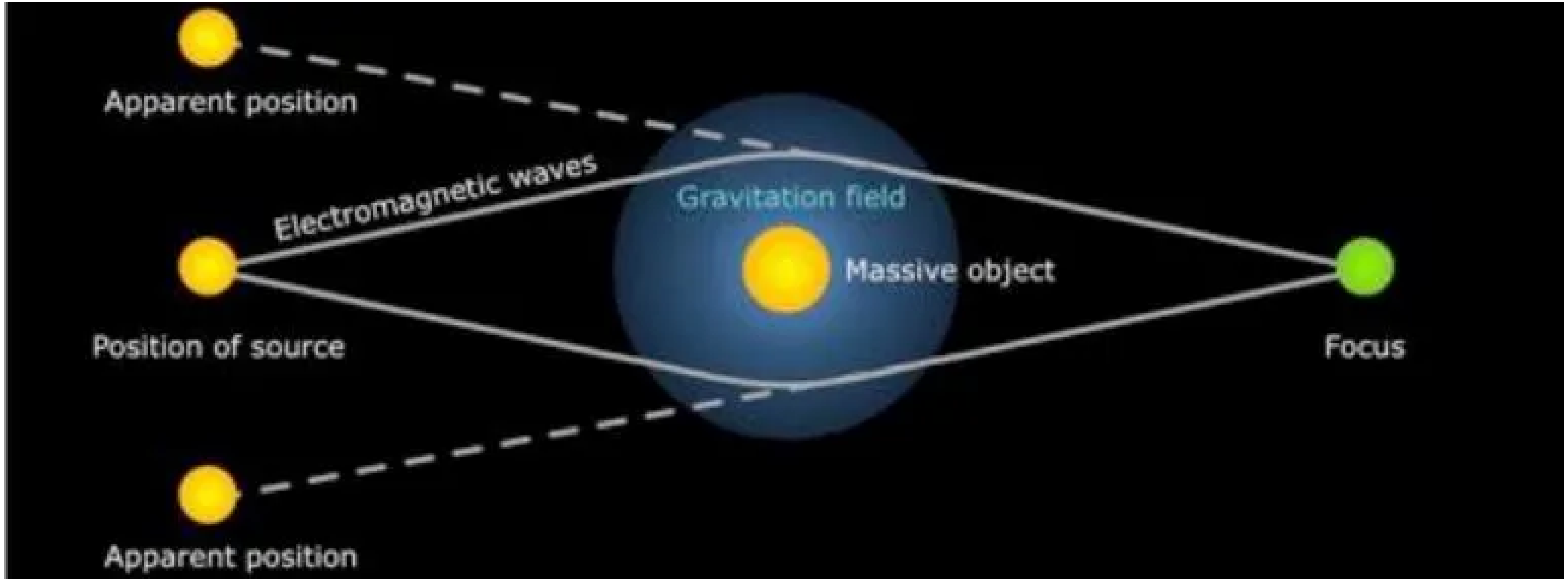
$d\tau = \sqrt{1 - r_s/r}\, dt$ (GPS).



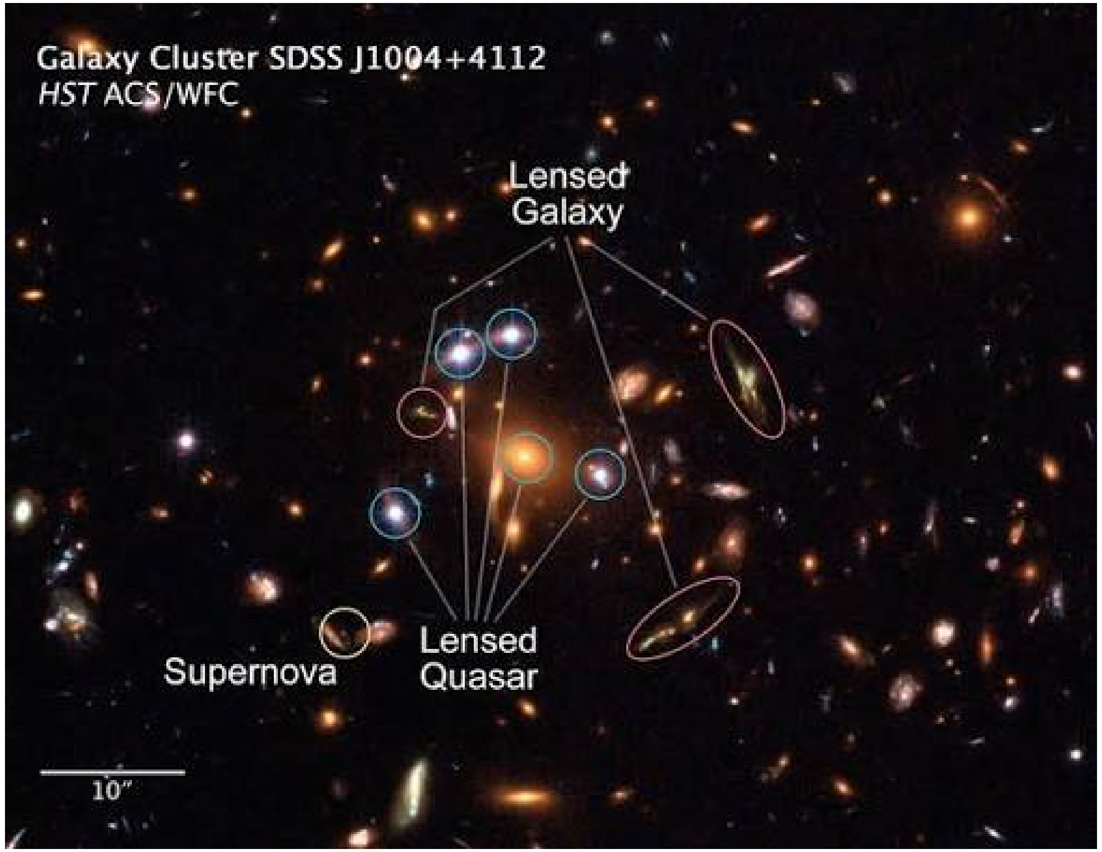
light cones tilt inward（光円錐）: at $r = r_s$ outgoing light runs in place; inside, all paths end at $r = 0$

(red: infalling, blue: outgoing light).

Gravity as a lens （重力レンズ）



Lensed images（重力レンズ像） in a real cluster

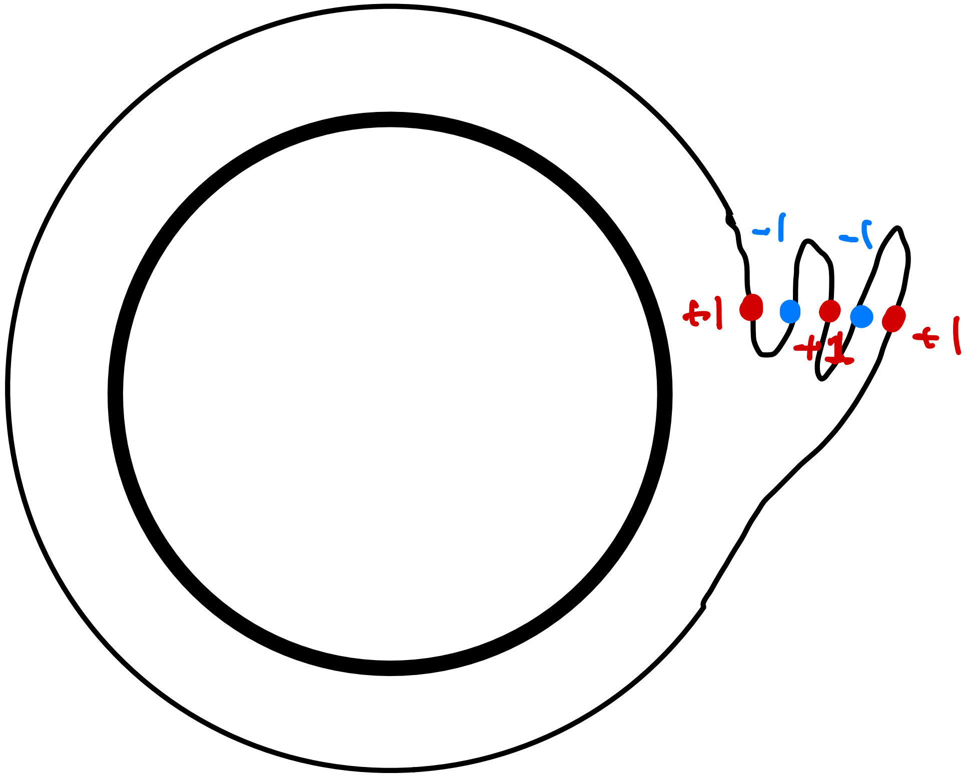


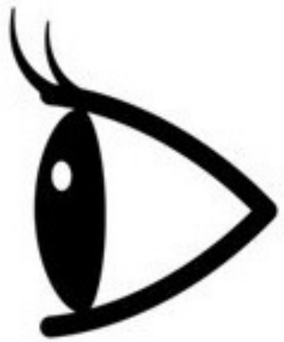
Galaxy cluster SDSS J1004+4112 (HST ACS/WFC)

FACT. The number of gravitational images (if no image is blocked) is always odd — and just less than half of them are *inverted*（反転） (mirror) images.

Degree of a map （写像度）

Net number of pre-images of a given point, counted with a "$+$" sign if the map is not inverted and a "$-$" sign if it is.





FACT. The number of gravitational images (if no image is blocked) is always odd — and just less than half of them are *inverted* (mirror) images.

SECTION 6

Summary and Outlook

まとめと展望

Physics	**Mathematics**
springs & harmonic oscillators ばねと調和振動子	eigenvectors, Fourier analysis 固有ベクトル，フーリエ解析
monopoles & charge quantization 磁気単極子と電荷の量子化	Hopf bundle, Chern number ホップ束，チャーン数
Biot–Savart law ビオ・サバールの法則	Gauss linking number, topology ガウスの絡み数，位相幾何学
quantum spin 量子スピン	vectors in $\mathbb{C}^2$, Hermitian matrices, $SU(2)$ $\mathbb{C}^2$ のベクトル，エルミート行列
time dilation & length contraction 時間の遅れと長さの収縮	Minkowski space, Lorentz transformation ミンコフスキー空間，ローレンツ変換
gravity 重力	curvature, geodesics 曲率，測地線
gravitational lens 重力レンズ	mapping degree, topology 写像度，位相幾何学

How to keep learning

- Spend time in the **library** and explore the wide range of physics and math books available.

  $\rightarrow$ Recommended Reading

- Regularly browse **mathematical magazines** such as 「数学セミナー」「数学のたのしみ」「数理科学」「数学」

- Go beyond your university courses by watching **online lectures** on topics that interest you, including those on **YouTube**.

- When you encounter something you do not understand, use **AI** as a tool to ask questions and clarify concepts.

- Most importantly, learn actively: work through problems yourself, write things down, and carry out calculations by hand.

あなたが気に入った方法，分野はきっとあなたに適しているのでしょう．ところで，あなたは すでに数学を志した．それは数学が好きになっ たからだ．数学のどんな所が好きになったのかよく反省して，数学のそのような所，方法，分野の発展した所を探せばよいかも知れない。しかし，数学は広いから，あなたのまだ知らぬ所に，まだまだあなたにピッタリのものがあるかも知れない．あまり早くから専門に固まらず，広い視野をもつことをお勧めします．

（久賀道郎／東京大学理学部数学科のオリエンテーションにて）

How simple are the laws of Nature?

Each fundamental theory is a single functional — the **action** S.

harmonic oscillator: $S = \int dt \left[\frac{m}{2}\dot{x}^2 - \frac{m\omega^2}{2}x^2\right]$

free particle (geodesic): $S = -mc \int ds$

QED: $S = \int d^4x \left[-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}\, i\gamma^\mu D_\mu \psi\right]$

QCD: $S = \int d^4x \left[-\frac{1}{2}\,\mathrm{tr}\, F_{\mu\nu}F^{\mu\nu} + \bar{q}\, i\gamma^\mu D_\mu q\right]$

Einstein–Hilbert: $S = \frac{c^4}{16\pi G}\int d^4x \sqrt{-g}\,(R - 2\Lambda)$